

Numerical and Experimental Study of Motion Control Using Pressure Feedback

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Abstract

This paper is concerned with the inherent oscillatory nature of pressure compensated velocity control of a hydraulic cylinder subjected to a negative load and suspended by means of a counter balance valve. A pressure feed back scheme that has as target to maintain the high pass filtered pressure gradient equal to zero is adopted with a view to reduce oscillations while maintaining other performance criteria related to the motion control. It yields lead compensation with a markedly improved performance. The sizing of the filter is described taking into account the bandwidth of the directional control valve. The suggested control scheme is implemented and examined in a non-linear time domain simulation model and a single degree-of-freedom test rig actuated by means of a double acting hydraulic cylinder.

Keywords: Counter balance valve, pressure feedback, negative load, experimental verification

1 Introduction

The use of over-center valves or counterbalance valves are widely used in hydraulics on such applications as cranes, telehandlers and winches as an integrated part in the actuator control. They are distributed to each actuator (degree of freedom) depending on the type of external loading. They are multi functional and, normally, they serve at least the following functions:

- leak tight load holding
- shock absorption
- cavitation protection at load lowering
- load holding at pipe burst
- no drop before lift

It is, however, well known that they tend to introduce instability in a system, especially when the flow supply is pressure compensated. This is the case when the directional control valve that supplies flow to the actuator suspended by the over center valve is equipped with an internal compensator in series with the main spool; a so-called flow control valve. As the latter is increasingly asked for by system manufacturers a major problem in present day hydraulics is to design pressure compensated systems containing over-center valves that offer stable load lowering performance without unnecessarily compromising system efficiency and response.

Naturally, this problem has attracted a lot of attention with emphasis on modeling and parameter variation [1..16] revealing a number of common stabilizing characteristics. They include increased volume between directional valve and actuator, reduced pilot area ratio as well as reduced pay

and inertia load, whereas the influence by a number of other design variables remain less obvious.

As pointed out in [7] the operation of the over-center valve is application sensitive, greatly increasing the complexity of choosing/designing it. Adding to the difficulties is the fact that hard to control quantities such as friction and hysteresis in the counter balance valve also are sources of instability [8]. Finally, it should be noted that stability only is one of several performance parameters for an over-center valve system. The different functionalities constitutes a set of design constraints whereas power consumption, stability, response time, load dependency, manufacturability and costs represents performance criteria.

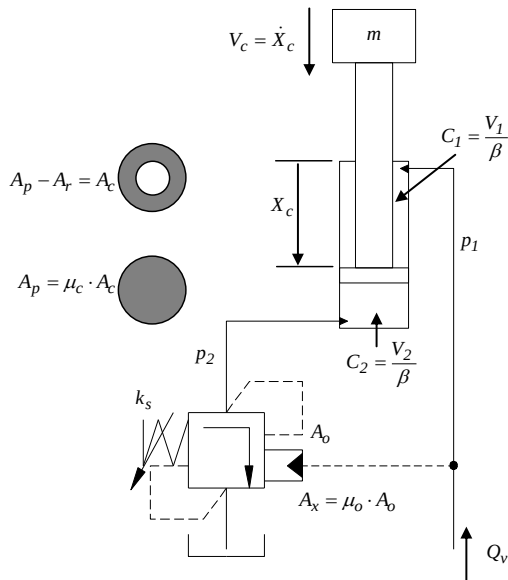
Therefore, this paper is concerned with the analysis and reduction of the oscillatory nature of an over center valve system taking into account that the above mentioned design criteria cannot be ignored. Recent work [16] focuses on closed-loop control. This is easily motivated by the increasing level of instrumentation and computational power in hydraulically actuated machinery today, however, the experimental implementation is needed. The main contribution of this paper is the practical implementation of the pressure feedback scheme put forward in [16] on a single degree-of-freedom test rig with nonlinear dynamic characteristics.

2 Considered system

In Figure 1 the main components of a pressure compensated over center valve circuit are shown. They comprise a 4/3 way directional control valve (in lowering position) with a compensator in series that maintains the pressure drop across the main spool metering orifice approximately

The diagram illustrates a hydraulic circuit for a payload. At the top, a cylinder is connected to a horizontal bar labeled "Payload". The cylinder's rod extends to the right. Below the cylinder, a 4/3-way directional control valve is shown. This valve has two main ports: one connected to the pump and another connected to the cylinder. The valve also has two return ports to the tank. The valve is controlled by an "Over-center valve" (a solenoid) and a "Compensator". The circuit includes a pump, a check valve, and a pressure compensator. The system is designed to maintain pressure in the cylinder when the payload is moved.

In fig. 2 a simplified circuit is shown including the core components and parameters.



The governing equations for this system are:

Thus, a lead compensation has been added. Choosing K_f substantially larger than C_1 gives a reduction of the gain of the open loop transfer function below ω_f while adding phase around ω_f . Since the phase lag of the open loop transfer function of the original system reaches 180° at ω_n ,

it would be obvious to position the filter frequency in the vicinity of ω_n . In that case the gain may be computed as:

$$K_f = \frac{\mu_o}{\mu_c} \cdot C_2 \quad (8)$$

Frequency analysis and non-linear time domain analysis clearly support the performance enhancement obtained by the proposed method. In [16] numerical studies are carried out that may be interpreted as follows: the filter frequency must be chosen in accordance with the bandwidth of the directional control valve in such a way that a slow directional control valve calls for a high filter frequency and vice versa., see also figs. 3 and 4.

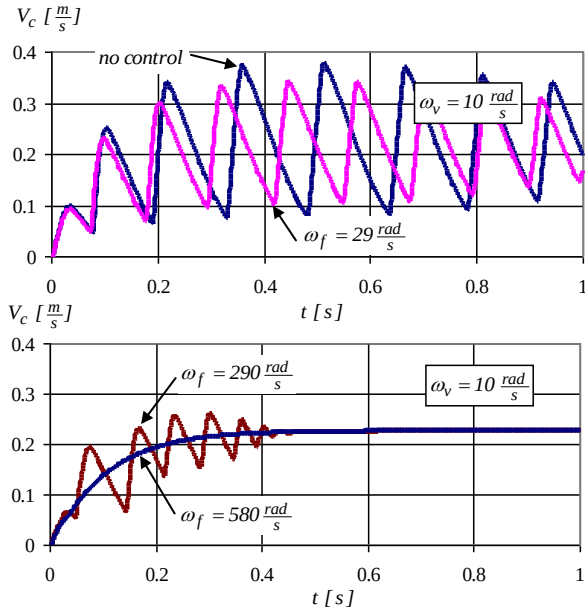


Figure 3: Piston velocity for $\omega_v = 10 \frac{\text{rad}}{\text{s}}$.

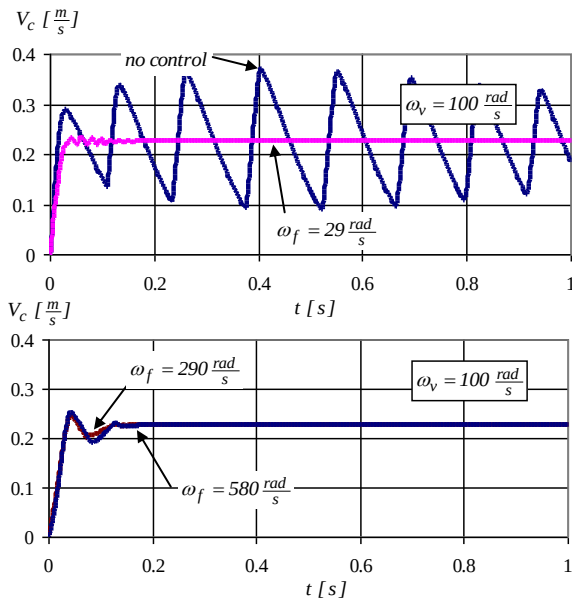


Figure 4: Piston velocity for $\omega_v = 100 \frac{\text{rad}}{\text{s}}$.

In this work these results are experimentally verified by means of the test rig shown in fig. 5

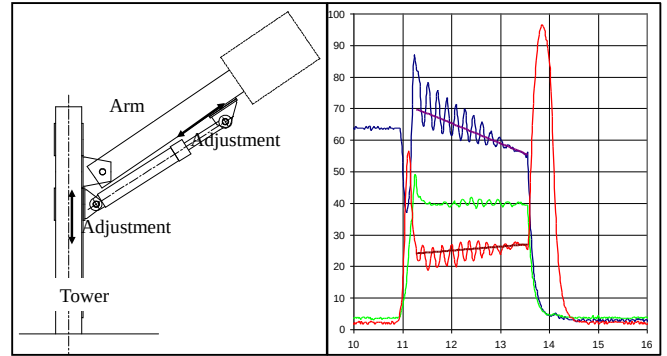


Figure 5: Simplified sketch of experimental test rig.

The main purpose of this work has been to evaluate the robustness, i.e., the ability of the feedback control scheme to work satisfactory under varying dynamic obtained by adjusting the points of attachment of the cylinder to the tower and the arm, see fig. 5.

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